APT Overlap: APT28 vs APT29 vs Sandworm Team (Parent + Sub-techniques)

Technique overlap across 3 selected groups | 129 parent T-codes + 140 sub-techniques | campaigns included | software-derived excluded

32 RED
techniques + 15 sub-techniques
Used by ALL 3 selected groups

31 YELLOW
techniques + 35 sub-techniques
Used by more than one, but not all

66 LIGHT GREEN
techniques + 90 sub-techniques
Used by only ONE selected group

Selected Groups

ID	Group	Aliases	T-codes
G0007	**APT28**	IRON TWILIGHT, SNAKEMACKEREL, Swallowtail, Group 74, Sednit, Sofacy, Pawn Storm, Fancy Bear, STRONTIUM, Tsar Team, Threat Group-4127, TG-4127, Forest Blizzard, FROZENLAKE, GruesomeLarch	77
G0016	**APT29**	IRON RITUAL, IRON HEMLOCK, NobleBaron, Dark Halo, NOBELIUM, UNC2452, YTTRIUM, The Dukes, Cozy Bear, CozyDuke, SolarStorm, Blue Kitsune, UNC3524, Midnight Blizzard	72
G0034	**Sandworm Team**	ELECTRUM, Telebots, IRON VIKING, BlackEnergy (Group), Quedagh, Voodoo Bear, IRIDIUM, Seashell Blizzard, FROZENBARENTS, APT44	75

Coverage by Tactic

Tactic	RED	YELLOW	LIGHT GREEN	Total
Reconnaissance	2	2	5	**9**
Resource Development	4	2	1	**7**
Initial Access	5	1	3	**9**
Execution	3	2	4	**9**
Persistence	4	5	6	**15**
Privilege Escalation	2	6	4	**12**
Stealth	6	0	9	**15**
Defense Impairment	1	2	3	**6**
Credential Access	2	5	7	**14**
Discovery	1	5	7	**13**
Lateral Movement	1	1	4	**6**
Collection	2	4	5	**11**
Command and Control	4	2	8	**14**
Exfiltration	0	1	3	**4**
Impact	0	1	7	**8**

Technique Matrix (1/2)

indented rows = sub-techniques, each colored by its own overlap

RED (32 + 15 sub) YELLOW (31 + 35 sub) LIGHT GREEN (66 + 90 sub)

Reconnaissance
9 techniques

- **T1589** 3/3 Gather Victim Identity Information
  - .001 Credentials 2/3
  - .002 Email Addres... 1/3
  - .003 Employee Na... 1/3
- **T1595** 3/3 Active Scanning
  - .002 Vulnerability ... 3/3
- **T1591** 2/3 Gather Victim Org Information
  - .002 Business Rel... 1/3
- **T1598** 2/3 Phishing for Information
  - .003 Spearphishin... 2/3
- **T1590** 1/3 Gather Victim Network Information
  - .001 Domain Prop... 1/3
- **T1592** 1/3 Gather Victim Host Information
  - .002 Software 1/3
- **T1593** 1/3 Search Open Websites/Domains
- **T1594** 1/3 Search Victim-Owned Websites
- **T1596** 1/3 Search Open Technical Databases

Resource Development
7 techniques

- **T1583** 3/3 Acquire Infrastructure
  - .001 Domains 3/3
  - .006 Web Services 2/3
  - .003 Virtual Privat... 1/3
  - .004 Server 1/3
- **T1584** 3/3 Compromise Infrastructure
  - .001 Domains 1/3
  - .004 Server 1/3
  - .005 Botnet 1/3
  - .008 Network Devi... 1/3
- **T1586** 3/3 Compromise Accounts
  - .002 Email Accounts 2/3
  - .001 Social Media ... 1/3
  - .003 Cloud Accou... 1/3
- **T1588** 3/3 Obtain Capabilities
  - .002 Tool 3/3
  - .006 Vulnerabilities 1/3
  - .007 Artificial Intelli... 1/3
- **T1585** 2/3 Establish Accounts
  - .001 Social Media ... 2/3
  - .002 Email Accounts 1/3
- **T1587** 2/3 Develop Capabilities
  - .001 Malware 2/3
  - .003 Digital Certific... 1/3
- **T1608** 1/3 Stage Capabilities
  - .001 Upload Malw... 1/3

Initial Access
9 techniques

- **T1078** 3/3 Valid Accounts
  - .002 Domain Acco... 2/3
  - .004 Cloud Accou... 2/3
  - .003 Local Accounts 1/3
- **T1133** 3/3 External Remote Services
- **T1190** 3/3 Exploit Public-Facing Application
- **T1199** 3/3 Trusted Relationship
- **T1566** 3/3 Phishing
  - .001 Spearphishin... 3/3
  - .002 Spearphishin... 2/3
  - .003 Spearphishin... 1/3
- **T1195** 2/3 Supply Chain Compromise
  - .002 Compromise ... 2/3
- **T1091** 1/3 Replication Through Removable Media
- **T1189** 1/3 Drive-by Compromise
- **T1669** 1/3 Wi-Fi Networks

Execution
9 techniques

- **T1059** 3/3 Command and Scripting Interpreter
  - .001 PowerShell 3/3
  - .003 Windows Co... 3/3
  - .005 Visual Basic 2/3
  - .006 Python 1/3
  - .009 Cloud API 1/3
- **T1203** 3/3 Exploitation for Client Execution
- **T1204** 3/3 User Execution
  - .001 Malicious Link 3/3
  - .002 Malicious File 3/3
- **T1047** 2/3 Windows Management Instrumentation
- **T1053** 2/3 Scheduled Task/Job
  - .005 Scheduled T... 2/3
- **T1072** 1/3 Software Deployment Tools
- **T1106** 1/3 Native API
- **T1559** 1/3 Inter-Process Communication
  - .002 Dynamic Dat... 1/3
- **T1651** 1/3 Cloud Administration Command

Persistence
15 techniques

- **T1078** 3/3 Valid Accounts
  - .002 Domain Acco... 2/3
  - .004 Cloud Accou... 2/3
  - .003 Local Accounts 1/3
- **T1098** 3/3 Account Manipulation
  - .002 Additional E... 2/3
  - .001 Additional Cl... 1/3
  - .003 Additional Cl... 1/3
  - .005 Device Regist... 1/3
- **T1133** 3/3 External Remote Services
- **T1505** 3/3 Server Software Component
  - .003 Web Shell 3/3
  - .001 SQL Stored ... 1/3
- **T1037** 2/3 Boot or Logon Initialization Scripts
  - .001 Logon Script ... 1/3
  - .004 RC Scripts 1/3
- **T1053** 2/3 Scheduled Task/Job
  - .005 Scheduled T... 2/3
- **T1136** 2/3 Create Account
  - .002 Domain Acco... 1/3
  - .003 Cloud Account 1/3
- **T1546** 2/3 Event Triggered Execution
  - .003 Windows Ma... 1/3
  - .008 Accessibility ... 1/3
  - .015 Component ... 1/3
- **T1547** 2/3 Boot or Logon Autostart Execution
  - .001 Registry Run ... 2/3
- **T1112** 1/3 Modify Registry
- **T1137** 1/3 Office Application Startup
  - .002 Office Test 1/3
- **T1542** 1/3 Pre-OS Boot
  - .003 Bootkit 1/3

Privilege Escalation
12 techniques

- **T1078** 3/3 Valid Accounts
  - .002 Domain Acco... 2/3
  - .004 Cloud Accou... 2/3
  - .003 Local Accounts 1/3
- **T1098** 3/3 Account Manipulation
  - .002 Additional E... 2/3
  - .001 Additional Cl... 1/3
  - .003 Additional Cl... 1/3
  - .005 Device Regist... 1/3
- **T1037** 2/3 Boot or Logon Initialization Scripts
  - .001 Logon Script ... 1/3
  - .004 RC Scripts 1/3
- **T1053** 2/3 Scheduled Task/Job
  - .005 Scheduled T... 2/3
- **T1068** 2/3 Exploitation for Privilege Escalation
- **T1484** 2/3 Domain or Tenant Policy Modification
  - .001 Group Policy ... 1/3
  - .002 Trust Modific... 1/3
- **T1546** 2/3 Event Triggered Execution
  - .003 Windows Ma... 1/3
  - .008 Accessibility ... 1/3
  - .015 Component ... 1/3
- **T1547** 2/3 Boot or Logon Autostart Execution
  - .001 Registry Run ... 2/3
- **T1055** 1/3 Process Injection
- **T1134** 1/3 Access Token Manipulation
  - .001 Token Impers... 1/3
- **T1543** 1/3 Create or Modify System Process
  - .002 Systemd Ser... 1/3
  - .003 Windows Ser... 1/3
- **T1548** 1/3 Abuse Elevation Control Mechanism
  - .002 Bypass User ... 1/3

Stealth
15 techniques

- **T1027** 3/3 Obfuscated Files or Information
  - .002 Software Pac... 2/3
  - .001 Binary Padding 1/3
  - .003 Steganography 1/3
  - .006 HTML Smugg... 1/3
  - .010 Command O... 1/3
  - .013 Encrypted/En... 1/3
- **T1036** 3/3 Masquerading
  - .005 Match Legiti... 3/3
  - .004 Masquerade ... 2/3
  - .008 Masquerade ... 1/3
  - .010 Masquerade ... 1/3
- **T1070** 3/3 Indicator Removal
  - .004 File Deletion 3/3
  - .006 Timestomp 2/3
  - .008 Clear Mailbox... 1/3
- **T1078** 3/3 Valid Accounts
  - .002 Domain Acco... 2/3
  - .004 Cloud Accou... 2/3
  - .003 Local Accounts 1/3
- **T1140** 3/3 Deobfuscate/Decode Files or Information
- **T1218** 3/3 System Binary Proxy Execution
  - .011 Rundll32 3/3
  - .005 Mshta 1/3
- **T1006** 1/3 Direct Volume Access
- **T1014** 1/3 Rootkit
- **T1055** 1/3 Process Injection
- **T1134** 1/3 Access Token Manipulation
  - .001 Token Impers... 1/3
- **T1211** 1/3 Exploitation for Stealth
- **T1221** 1/3 Template Injection
- **T1542** 1/3 Pre-OS Boot
  - .003 Bootkit 1/3

Defense Impairment
6 techniques

- **T1685** 3/3 Disable or Modify Tools
  - .001 Disable or Mo... 2/3
  - .002 Disable or Mo... 1/3
  - .005 Clear Windo... 1/3
- **T1484** 2/3 Domain or Tenant Policy Modification
  - .001 Group Policy ... 1/3
  - .002 Trust Modific... 1/3
- **T1686** 2/3 Disable or Modify System Firewall
  - .003 Windows Hos... 1/3
- **T1112** 1/3 Modify Registry
- **T1553** 1/3 Subvert Trust Controls
  - .002 Code Signing 1/3
  - .005 Mark-of-the-... 1/3
- **T1556** 1/3 Modify Authentication Process
  - .007 Hybrid Identity 1/3

Credential Access
14 techniques

- **T1003** 3/3 OS Credential Dumping
  - .001 LSASS Mem... 2/3
  - .002 Security Acco... 2/3
  - .003 NTDS 2/3
  - .004 LSA Secrets 1/3
  - .006 DCSync 1/3
- **T1110** 3/3 Brute Force
  - .001 Password Gu... 2/3
  - .003 Password Sp... 2/3
- **T1040** 2/3 Network Sniffing
- **T1056** 2/3 Input Capture
  - .001 Keylogging 2/3
- **T1528** 2/3 Steal Application Access Token
- **T1539** 2/3 Steal Web Session Cookie
- **T1555** 2/3 Credentials from Password Stores
  - .003 Credentials fr... 2/3
- **T1552** 1/3 Unsecured Credentials
  - .004 Private Keys 1/3
- **T1556** 1/3 Modify Authentication Process
  - .007 Hybrid Identity 1/3
- **T1557** 1/3 Adversary-in-the-Middle
  - .004 Evil Twin 1/3
- **T1558** 1/3 Steal or Forge Kerberos Tickets
  - .003 Kerberoasting 1/3
- **T1606** 1/3 Forge Web Credentials
  - .001 Web Cookies 1/3
  - .002 SAML Tokens 1/3
- **T1621** 1/3 Multi-Factor Authentication Reque...
- **T1649** 1/3 Steal or Forge Authentication Certific...

Discovery
13 techniques

- **T1083** 3/3 File and Directory Discovery
- **T1016** 2/3 System Network Configuration Discovery
  - .001 Internet Conn... 1/3
  - .002 Wi-Fi Discovery 1/3
- **T1018** 2/3 Remote System Discovery
- **T1040** 2/3 Network Sniffing
- **T1057** 2/3 Process Discovery
- **T1087** 2/3 Account Discovery
  - .002 Domain Acco... 2/3
  - .003 Email Account 1/3
  - .004 Cloud Account 1/3
- **T1033** 1/3 System Owner/User Discovery
- **T1049** 1/3 System Network Connections Discovery
- **T1069** 1/3 Permission Groups Discovery
  - .002 Domain Groups 1/3
- **T1082** 1/3 System Information Discovery
- **T1120** 1/3 Peripheral Device Discovery
- **T1482** 1/3 Domain Trust Discovery
- **T1680** 1/3 Local Storage Discovery

Lateral Movement
6 techniques

- **T1021** 3/3 Remote Services
  - .002 SMB/Window... 3/3
  - .001 Remote Desk... 2/3
  - .006 Windows Re... 1/3
  - .007 Cloud Services 1/3
- **T1550** 2/3 Use Alternate Authentication Material
  - .001 Application A... 2/3
  - .002 Pass the Hash 1/3
  - .003 Pass the Ticket 1/3
  - .004 Web Session... 1/3
- **T1072** 1/3 Software Deployment Tools
- **T1091** 1/3 Replication Through Removable Media
- **T1210** 1/3 Exploitation of Remote Services
- **T1570** 1/3 Lateral Tool Transfer

Collection
11 techniques

- **T1005** 3/3 Data from Local System
- **T1213** 3/3 Data from Information Repositories
  - .002 Sharepoint 1/3
  - .003 Code Reposit... 1/3
  - .006 Databases 1/3
- **T1056** 2/3 Input Capture
  - .001 Keylogging 2/3
- **T1074** 2/3 Data Staged
  - .002 Remote Data... 2/3
  - .001 Local Data St... 1/3
- **T1114** 2/3 Email Collection
  - .002 Remote Emai... 2/3
- **T1560** 2/3 Archive Collected Data
  - .001 Archive via Ut... 2/3
- **T1025** 1/3 Data from Removable Media
- **T1039** 1/3 Data from Network Shared Drive
- **T1113** 1/3 Screen Capture
- **T1119** 1/3 Automated Collection
- **T1557** 1/3 Adversary-in-the-Middle
  - .004 Evil Twin 1/3

Command and Control
14 techniques

- **T1071** 3/3 Application Layer Protocol
  - .001 Web Protocols 3/3
  - .003 Mail Protocols 1/3
- **T1090** 3/3 Proxy
  - .001 Internal Proxy 2/3
  - .002 External Proxy 2/3
  - .003 Multi-hop Proxy 2/3
  - .004 Domain Front... 1/3
- **T1102** 3/3 Web Service
  - .002 Bidirectional ... 3/3
- **T1105** 3/3 Ingress Tool Transfer
- **T1001** 2/3 Data Obfuscation
  - .001 Junk Data 1/3
  - .002 Steganography 1/3
- **T1573** 2/3 Encrypted Channel
  - .001 Symmetric Cr... 1/3
- **T1092** 1/3 Communication Through Removable ...
- **T1095** 1/3 Non-Application Layer Protocol
- **T1132** 1/3 Data Encoding
  - .001 Standard Enc... 1/3
- **T1219** 1/3 Remote Access Tools
- **T1568** 1/3 Dynamic Resolution
- **T1571** 1/3 Non-Standard Port
- **T1572** 1/3 Protocol Tunneling
- **T1665** 1/3 Hide Infrastructure

Exfiltration
4 techniques

- **T1048** 2/3 Exfiltration Over Alternative Protocol
  - .002 Exfiltration O... 2/3
- **T1030** 1/3 Data Transfer Size Limits
- **T1041** 1/3 Exfiltration Over C2 Channel
- **T1567** 1/3 Exfiltration Over Web Service

Impact
8 techniques

- **T1561** 2/3 Disk Wipe
  - .001 Disk Content ... 1/3
  - .002 Disk Structur... 1/3
- **T1485** 1/3 Data Destruction
- **T1486** 1/3 Data Encrypted for Impact
- **T1489** 1/3 Service Stop
- **T1490** 1/3 Inhibit System Recovery
- **T1491** 1/3 Defacement
  - .002 External Defa... 1/3
- **T1498** 1/3 Network Denial of Service
- **T1499** 1/3 Endpoint Denial of Service

Technique Matrix (2/2)

indented rows = sub-techniques, each colored by its own overlap

RED (32 + 15 sub) YELLOW (31 + 35 sub) LIGHT GREEN (66 + 90 sub)

Reconnaissance	Resource Development	Initial Access	Execution	Persistence	Privilege Escalation	Stealth	Defense Impairment	Credential Access	Discovery	Lateral Movement	Collection	Command and Control	Exfiltration	Impact
9 techniques	7 techniques	9 techniques	9 techniques	15 techniques	12 techniques	15 techniques	6 techniques	14 techniques	13 techniques	6 techniques	11 techniques	14 techniques	4 techniques	8 techniques
				T1543 1/3 Create or Modify System Process		T1564 1/3 Hide Artifacts								
				.002 Systemd Ser... 1/3		.001 Hidden Files ... 1/3								
				.003 Windows Ser... 1/3		.003 Hidden Window 1/3								
				T1554 1/3 Compromise Host Software Binary		T1684 1/3 Social Engineering								
				T1556 1/3 Modify Authentication Process		.001 Impersonation 1/3								
				.007 Hybrid Identity 1/3										

Technique & Sub-Technique Detail

Overlap	Groups	ID	Technique / Sub-technique	Tactics	APT28	APT29	Sandworm Team
RED	3/3	T1003	**OS Credential Dumping**	Credential Access	✔	✔	✔
YELLOW	2/3	T1003.001	LSASS Memory		✔		✔
YELLOW	2/3	T1003.002	Security Account Manager		✔	✔	
YELLOW	2/3	T1003.003	NTDS		✔		✔
LIGHT GREEN	1/3	T1003.004	LSA Secrets			✔	
LIGHT GREEN	1/3	T1003.006	DCSync			✔	
RED	3/3	T1005	**Data from Local System**	Collection	✔	✔	✔
RED	3/3	T1021	**Remote Services**	Lateral Movement	✔	✔	✔
RED	3/3	T1021.002	SMB/Windows Admin Shares		✔	✔	✔
YELLOW	2/3	T1021.001	Remote Desktop Protocol		✔	✔	
LIGHT GREEN	1/3	T1021.006	Windows Remote Management			✔	
LIGHT GREEN	1/3	T1021.007	Cloud Services			✔	
RED	3/3	T1027	**Obfuscated Files or Information**	Stealth	✔	✔	✔
YELLOW	2/3	T1027.002	Software Packing			✔	✔
LIGHT GREEN	1/3	T1027.001	Binary Padding			✔	
LIGHT GREEN	1/3	T1027.003	Steganography			✔	
LIGHT GREEN	1/3	T1027.006	HTML Smuggling			✔	
LIGHT GREEN	1/3	T1027.010	Command Obfuscation				✔
LIGHT GREEN	1/3	T1027.013	Encrypted/Encoded File		✔		
RED	3/3	T1036	**Masquerading**	Stealth	✔	✔	✔
RED	3/3	T1036.005	Match Legitimate Resource Name or Location		✔	✔	✔
YELLOW	2/3	T1036.004	Masquerade Task or Service			✔	✔
LIGHT GREEN	1/3	T1036.008	Masquerade File Type				✔
LIGHT GREEN	1/3	T1036.010	Masquerade Account Name				✔
RED	3/3	T1059	**Command and Scripting Interpreter**	Execution	✔	✔	✔
RED	3/3	T1059.001	PowerShell		✔	✔	✔
RED	3/3	T1059.003	Windows Command Shell		✔	✔	✔
YELLOW	2/3	T1059.005	Visual Basic			✔	✔
LIGHT GREEN	1/3	T1059.006	Python			✔	
LIGHT GREEN	1/3	T1059.009	Cloud API			✔	
RED	3/3	T1070	**Indicator Removal**	Stealth	✔	✔	✔
RED	3/3	T1070.004	File Deletion		✔	✔	✔
YELLOW	2/3	T1070.006	Timestomp		✔	✔	
LIGHT GREEN	1/3	T1070.008	Clear Mailbox Data			✔	
RED	3/3	T1071	**Application Layer Protocol**	Command and Control	✔	✔	✔
RED	3/3	T1071.001	Web Protocols		✔	✔	✔
LIGHT GREEN	1/3	T1071.003	Mail Protocols		✔		
RED	3/3	T1078	**Valid Accounts**	Initial Access, Persistence, Privilege Escalation, Stealth	✔	✔	✔
YELLOW	2/3	T1078.002	Domain Accounts			✔	✔
YELLOW	2/3	T1078.004	Cloud Accounts		✔	✔	
LIGHT GREEN	1/3	T1078.003	Local Accounts			✔	
RED	3/3	T1083	**File and Directory Discovery**	Discovery	✔	✔	✔
RED	3/3	T1090	**Proxy**	Command and Control	✔	✔	✔

Overlap	Groups	ID	Technique / Sub-technique	Tactics	APT28	APT29	Sandworm Team
YELLOW	2/3	T1090.001	Internal Proxy		✔	✔	
YELLOW	2/3	T1090.002	External Proxy		✔	✔	
YELLOW	2/3	T1090.003	Multi-hop Proxy		✔	✔	
LIGHT GREEN	1/3	T1090.004	Domain Fronting			✔	
RED	3/3	T1098	**Account Manipulation**	Persistence, Privilege Escalation	✔	✔	✔
YELLOW	2/3	T1098.002	Additional Email Delegate Permissions		✔	✔	
LIGHT GREEN	1/3	T1098.001	Additional Cloud Credentials			✔	
LIGHT GREEN	1/3	T1098.003	Additional Cloud Roles			✔	
LIGHT GREEN	1/3	T1098.005	Device Registration			✔	
RED	3/3	T1102	**Web Service**	Command and Control	✔	✔	✔
RED	3/3	T1102.002	Bidirectional Communication		✔	✔	✔
RED	3/3	T1105	**Ingress Tool Transfer**	Command and Control	✔	✔	✔
RED	3/3	T1110	**Brute Force**	Credential Access	✔	✔	✔
YELLOW	2/3	T1110.001	Password Guessing		✔	✔	
YELLOW	2/3	T1110.003	Password Spraying		✔	✔	
RED	3/3	T1133	**External Remote Services**	Initial Access, Persistence	✔	✔	✔
RED	3/3	T1140	**Deobfuscate/Decode Files or Information**	Stealth	✔	✔	✔
RED	3/3	T1190	**Exploit Public-Facing Application**	Initial Access	✔	✔	✔
RED	3/3	T1199	**Trusted Relationship**	Initial Access	✔	✔	✔
RED	3/3	T1203	**Exploitation for Client Execution**	Execution	✔	✔	✔
RED	3/3	T1204	**User Execution**	Execution	✔	✔	✔
RED	3/3	T1204.001	Malicious Link		✔	✔	✔
RED	3/3	T1204.002	Malicious File		✔	✔	✔
RED	3/3	T1213	**Data from Information Repositories**	Collection	✔	✔	✔
LIGHT GREEN	1/3	T1213.002	Sharepoint		✔		
LIGHT GREEN	1/3	T1213.003	Code Repositories			✔	
LIGHT GREEN	1/3	T1213.006	Databases				✔
RED	3/3	T1218	**System Binary Proxy Execution**	Stealth	✔	✔	✔
RED	3/3	T1218.011	Rundll32		✔	✔	✔
LIGHT GREEN	1/3	T1218.005	Mshta			✔	
RED	3/3	T1505	**Server Software Component**	Persistence	✔	✔	✔
RED	3/3	T1505.003	Web Shell		✔	✔	✔
LIGHT GREEN	1/3	T1505.001	SQL Stored Procedures				✔
RED	3/3	T1566	**Phishing**	Initial Access	✔	✔	✔
RED	3/3	T1566.001	Spearphishing Attachment		✔	✔	✔
YELLOW	2/3	T1566.002	Spearphishing Link			✔	✔
LIGHT GREEN	1/3	T1566.003	Spearphishing via Service			✔	
RED	3/3	T1583	**Acquire Infrastructure**	Resource Development	✔	✔	✔
RED	3/3	T1583.001	Domains		✔	✔	✔
YELLOW	2/3	T1583.006	Web Services		✔	✔	
LIGHT GREEN	1/3	T1583.003	Virtual Private Server		✔		
LIGHT GREEN	1/3	T1583.004	Server				✔
RED	3/3	T1584	**Compromise Infrastructure**	Resource Development	✔	✔	✔
LIGHT GREEN	1/3	T1584.001	Domains			✔	
LIGHT GREEN	1/3	T1584.004	Server				✔

Overlap	Groups	ID	Technique / Sub-technique	Tactics	APT28	APT29	Sandworm Team
LIGHT GREEN	1/3	T1584.005	Botnet				✔
LIGHT GREEN	1/3	T1584.008	Network Devices		✔		
RED	3/3	T1586	**Compromise Accounts**	Resource Development	✔	✔	✔
YELLOW	2/3	T1586.002	Email Accounts		✔	✔	
LIGHT GREEN	1/3	T1586.001	Social Media Accounts				✔
LIGHT GREEN	1/3	T1586.003	Cloud Accounts			✔	
RED	3/3	T1588	**Obtain Capabilities**	Resource Development	✔	✔	✔
RED	3/3	T1588.002	Tool		✔	✔	✔
LIGHT GREEN	1/3	T1588.006	Vulnerabilities				✔
LIGHT GREEN	1/3	T1588.007	Artificial Intelligence		✔		
RED	3/3	T1589	**Gather Victim Identity Information**	Reconnaissance	✔	✔	✔
YELLOW	2/3	T1589.001	Credentials		✔	✔	
LIGHT GREEN	1/3	T1589.002	Email Addresses				✔
LIGHT GREEN	1/3	T1589.003	Employee Names				✔
RED	3/3	T1595	**Active Scanning**	Reconnaissance	✔	✔	✔
RED	3/3	T1595.002	Vulnerability Scanning		✔	✔	✔
RED	3/3	T1685	**Disable or Modify Tools**	Defense Impairment	✔	✔	✔
YELLOW	2/3	T1685.001	Disable or Modify Windows Event Log			✔	✔
LIGHT GREEN	1/3	T1685.002	Disable or Modify Cloud Log			✔	
LIGHT GREEN	1/3	T1685.005	Clear Windows Event Logs		✔		
YELLOW	2/3	T1001	**Data Obfuscation**	Command and Control	✔	✔	
LIGHT GREEN	1/3	T1001.001	Junk Data		✔		
LIGHT GREEN	1/3	T1001.002	Steganography			✔	
YELLOW	2/3	T1016	**System Network Configuration Discovery**	Discovery	✔	✔	
LIGHT GREEN	1/3	T1016.001	Internet Connection Discovery			✔	
LIGHT GREEN	1/3	T1016.002	Wi-Fi Discovery		✔		
YELLOW	2/3	T1018	**Remote System Discovery**	Discovery		✔	✔
YELLOW	2/3	T1037	**Boot or Logon Initialization Scripts**	Persistence, Privilege Escalation	✔	✔	
LIGHT GREEN	1/3	T1037.001	Logon Script (Windows)		✔		
LIGHT GREEN	1/3	T1037.004	RC Scripts			✔	
YELLOW	2/3	T1040	**Network Sniffing**	Credential Access, Discovery	✔		✔
YELLOW	2/3	T1047	**Windows Management Instrumentation**	Execution		✔	✔
YELLOW	2/3	T1048	**Exfiltration Over Alternative Protocol**	Exfiltration	✔	✔	
YELLOW	2/3	T1048.002	Exfiltration Over Asymmetric Encrypted Non-C2 Protocol		✔	✔	
YELLOW	2/3	T1053	**Scheduled Task/Job**	Execution, Persistence, Privilege Escalation		✔	✔
YELLOW	2/3	T1053.005	Scheduled Task			✔	✔
YELLOW	2/3	T1056	**Input Capture**	Credential Access, Collection	✔		✔
YELLOW	2/3	T1056.001	Keylogging		✔		✔
YELLOW	2/3	T1057	**Process Discovery**	Discovery	✔	✔	
YELLOW	2/3	T1068	**Exploitation for Privilege Escalation**	Privilege Escalation	✔	✔	
YELLOW	2/3	T1074	**Data Staged**	Collection	✔	✔	
YELLOW	2/3	T1074.002	Remote Data Staging		✔	✔	
LIGHT GREEN	1/3	T1074.001	Local Data Staging		✔		

Overlap	Groups	ID	Technique / Sub-technique	Tactics	APT28	APT29	Sandworm Team
YELLOW	**2/3**	**T1087**	**Account Discovery**	Discovery		✔	✔
YELLOW	**2/3**	T1087.002	Domain Account			✔	✔
LIGHT GREEN	**1/3**	T1087.003	Email Account				✔
LIGHT GREEN	**1/3**	T1087.004	Cloud Account			✔	
YELLOW	**2/3**	**T1114**	**Email Collection**	Collection	✔	✔	
YELLOW	**2/3**	T1114.002	Remote Email Collection		✔	✔	
YELLOW	**2/3**	**T1136**	**Create Account**	Persistence		✔	✔
LIGHT GREEN	**1/3**	T1136.002	Domain Account				✔
LIGHT GREEN	**1/3**	T1136.003	Cloud Account			✔	
YELLOW	**2/3**	**T1195**	**Supply Chain Compromise**	Initial Access		✔	✔
YELLOW	**2/3**	T1195.002	Compromise Software Supply Chain			✔	✔
YELLOW	**2/3**	**T1484**	**Domain or Tenant Policy Modification**	Privilege Escalation, Defense Impairment		✔	✔
LIGHT GREEN	**1/3**	T1484.001	Group Policy Modification				✔
LIGHT GREEN	**1/3**	T1484.002	Trust Modification			✔	
YELLOW	**2/3**	**T1528**	**Steal Application Access Token**	Credential Access	✔	✔	
YELLOW	**2/3**	**T1539**	**Steal Web Session Cookie**	Credential Access		✔	✔
YELLOW	**2/3**	**T1546**	**Event Triggered Execution**	Persistence, Privilege Escalation	✔	✔	
LIGHT GREEN	**1/3**	T1546.003	Windows Management Instrumentation Event Subscription			✔	
LIGHT GREEN	**1/3**	T1546.008	Accessibility Features			✔	
LIGHT GREEN	**1/3**	T1546.015	Component Object Model Hijacking		✔		
YELLOW	**2/3**	**T1547**	**Boot or Logon Autostart Execution**	Persistence, Privilege Escalation	✔	✔	
YELLOW	**2/3**	T1547.001	Registry Run Keys / Startup Folder		✔	✔	
YELLOW	**2/3**	**T1550**	**Use Alternate Authentication Material**	Lateral Movement	✔	✔	
YELLOW	**2/3**	T1550.001	Application Access Token		✔	✔	
LIGHT GREEN	**1/3**	T1550.002	Pass the Hash		✔		
LIGHT GREEN	**1/3**	T1550.003	Pass the Ticket			✔	
LIGHT GREEN	**1/3**	T1550.004	Web Session Cookie			✔	
YELLOW	**2/3**	**T1555**	**Credentials from Password Stores**	Credential Access		✔	✔
YELLOW	**2/3**	T1555.003	Credentials from Web Browsers			✔	✔
YELLOW	**2/3**	**T1560**	**Archive Collected Data**	Collection	✔	✔	
YELLOW	**2/3**	T1560.001	Archive via Utility		✔	✔	
YELLOW	**2/3**	**T1561**	**Disk Wipe**	Impact	✔		✔
LIGHT GREEN	**1/3**	T1561.001	Disk Content Wipe		✔		
LIGHT GREEN	**1/3**	T1561.002	Disk Structure Wipe				✔
YELLOW	**2/3**	**T1573**	**Encrypted Channel**	Command and Control	✔	✔	
LIGHT GREEN	**1/3**	T1573.001	Symmetric Cryptography		✔		
YELLOW	**2/3**	**T1585**	**Establish Accounts**	Resource Development		✔	✔
YELLOW	**2/3**	T1585.001	Social Media Accounts			✔	✔
LIGHT GREEN	**1/3**	T1585.002	Email Accounts				✔
YELLOW	**2/3**	**T1587**	**Develop Capabilities**	Resource Development		✔	✔
YELLOW	**2/3**	T1587.001	Malware			✔	✔
LIGHT GREEN	**1/3**	T1587.003	Digital Certificates			✔	

Overlap	Groups	ID	Technique / Sub-technique	Tactics	APT28	APT29	Sandworm Team
YELLOW	2/3	T1591	Gather Victim Org Information	Reconnaissance	✔		✔
LIGHT GREEN	1/3	T1591.002	Business Relationships				✔
YELLOW	2/3	T1598	Phishing for Information	Reconnaissance	✔		✔
YELLOW	2/3	T1598.003	Spearphishing Link		✔		✔
YELLOW	2/3	T1686	Disable or Modify System Firewall	Defense Impairment	✔	✔	
LIGHT GREEN	1/3	T1686.003	Windows Host Firewall		✔		
LIGHT GREEN	1/3	T1006	Direct Volume Access	Stealth	✔		
LIGHT GREEN	1/3	T1014	Rootkit	Stealth	✔		
LIGHT GREEN	1/3	T1025	Data from Removable Media	Collection	✔		
LIGHT GREEN	1/3	T1030	Data Transfer Size Limits	Exfiltration	✔		
LIGHT GREEN	1/3	T1033	System Owner/User Discovery	Discovery			✔
LIGHT GREEN	1/3	T1039	Data from Network Shared Drive	Collection	✔		
LIGHT GREEN	1/3	T1041	Exfiltration Over C2 Channel	Exfiltration			✔
LIGHT GREEN	1/3	T1049	System Network Connections Discovery	Discovery			✔
LIGHT GREEN	1/3	T1055	Process Injection	Privilege Escalation, Stealth			✔
LIGHT GREEN	1/3	T1069	Permission Groups Discovery	Discovery		✔	
LIGHT GREEN	1/3	T1069.002	Domain Groups			✔	
LIGHT GREEN	1/3	T1072	Software Deployment Tools	Execution, Lateral Movement			✔
LIGHT GREEN	1/3	T1082	System Information Discovery	Discovery			✔
LIGHT GREEN	1/3	T1091	Replication Through Removable Media	Initial Access, Lateral Movement	✔		
LIGHT GREEN	1/3	T1092	Communication Through Removable Media	Command and Control	✔		
LIGHT GREEN	1/3	T1095	Non-Application Layer Protocol	Command and Control			✔
LIGHT GREEN	1/3	T1106	Native API	Execution			✔
LIGHT GREEN	1/3	T1112	Modify Registry	Persistence, Defense Impairment			✔
LIGHT GREEN	1/3	T1113	Screen Capture	Collection	✔		
LIGHT GREEN	1/3	T1119	Automated Collection	Collection	✔		
LIGHT GREEN	1/3	T1120	Peripheral Device Discovery	Discovery	✔		
LIGHT GREEN	1/3	T1132	Data Encoding	Command and Control			✔
LIGHT GREEN	1/3	T1132.001	Standard Encoding				✔
LIGHT GREEN	1/3	T1134	Access Token Manipulation	Privilege Escalation, Stealth	✔		
LIGHT GREEN	1/3	T1134.001	Token Impersonation/Theft		✔		
LIGHT GREEN	1/3	T1137	Office Application Startup	Persistence	✔		
LIGHT GREEN	1/3	T1137.002	Office Test		✔		
LIGHT GREEN	1/3	T1189	Drive-by Compromise	Initial Access	✔		
LIGHT GREEN	1/3	T1210	Exploitation of Remote Services	Lateral Movement	✔		
LIGHT GREEN	1/3	T1211	Exploitation for Stealth	Stealth	✔		
LIGHT GREEN	1/3	T1219	Remote Access Tools	Command and Control			✔
LIGHT GREEN	1/3	T1221	Template Injection	Stealth	✔		

Overlap	Groups	ID	Technique / Sub-technique	Tactics	APT28	APT29	Sandworm Team
LIGHT GREEN	1/3	T1482	Domain Trust Discovery	Discovery		✔	
LIGHT GREEN	1/3	T1485	Data Destruction	Impact			✔
LIGHT GREEN	1/3	T1486	Data Encrypted for Impact	Impact			✔
LIGHT GREEN	1/3	T1489	Service Stop	Impact			✔
LIGHT GREEN	1/3	T1490	Inhibit System Recovery	Impact			✔
LIGHT GREEN	1/3	T1491	Defacement	Impact			✔
LIGHT GREEN	1/3	T1491.002	External Defacement				✔
LIGHT GREEN	1/3	T1498	Network Denial of Service	Impact	✔		
LIGHT GREEN	1/3	T1499	Endpoint Denial of Service	Impact			✔
LIGHT GREEN	1/3	T1542	Pre-OS Boot	Persistence, Stealth	✔		
LIGHT GREEN	1/3	T1542.003	Bootkit		✔		
LIGHT GREEN	1/3	T1543	Create or Modify System Process	Persistence, Privilege Escalation			✔
LIGHT GREEN	1/3	T1543.002	Systemd Service				✔
LIGHT GREEN	1/3	T1543.003	Windows Service				✔
LIGHT GREEN	1/3	T1548	Abuse Elevation Control Mechanism	Privilege Escalation		✔	
LIGHT GREEN	1/3	T1548.002	Bypass User Account Control			✔	
LIGHT GREEN	1/3	T1552	Unsecured Credentials	Credential Access		✔	
LIGHT GREEN	1/3	T1552.004	Private Keys			✔	
LIGHT GREEN	1/3	T1553	Subvert Trust Controls	Defense Impairment		✔	
LIGHT GREEN	1/3	T1553.002	Code Signing			✔	
LIGHT GREEN	1/3	T1553.005	Mark-of-the-Web Bypass			✔	
LIGHT GREEN	1/3	T1554	Compromise Host Software Binary	Persistence			✔
LIGHT GREEN	1/3	T1556	Modify Authentication Process	Persistence, Defense Impairment, Credential Access		✔	
LIGHT GREEN	1/3	T1556.007	Hybrid Identity			✔	
LIGHT GREEN	1/3	T1557	Adversary-in-the-Middle	Credential Access, Collection	✔		
LIGHT GREEN	1/3	T1557.004	Evil Twin		✔		
LIGHT GREEN	1/3	T1558	Steal or Forge Kerberos Tickets	Credential Access		✔	
LIGHT GREEN	1/3	T1558.003	Kerberoasting			✔	
LIGHT GREEN	1/3	T1559	Inter-Process Communication	Execution	✔		
LIGHT GREEN	1/3	T1559.002	Dynamic Data Exchange		✔		
LIGHT GREEN	1/3	T1564	Hide Artifacts	Stealth	✔		
LIGHT GREEN	1/3	T1564.001	Hidden Files and Directories		✔		
LIGHT GREEN	1/3	T1564.003	Hidden Window		✔		
LIGHT GREEN	1/3	T1567	Exfiltration Over Web Service	Exfiltration	✔		
LIGHT GREEN	1/3	T1568	Dynamic Resolution	Command and Control		✔	
LIGHT GREEN	1/3	T1570	Lateral Tool Transfer	Lateral Movement			✔
LIGHT GREEN	1/3	T1571	Non-Standard Port	Command and Control			✔
LIGHT GREEN	1/3	T1572	Protocol Tunneling	Command and Control			✔
LIGHT GREEN	1/3	T1590	Gather Victim Network Information	Reconnaissance			✔

Overlap	Groups	ID	Technique / Sub-technique	Tactics	APT28	APT29	Sandworm Team
LIGHT GREEN	1/3	T1590.001	Domain Properties				✔
LIGHT GREEN	1/3	T1592	Gather Victim Host Information	Reconnaissance			✔
LIGHT GREEN	1/3	T1592.002	Software				✔
LIGHT GREEN	1/3	T1593	Search Open Websites/Domains	Reconnaissance			✔
LIGHT GREEN	1/3	T1594	Search Victim-Owned Websites	Reconnaissance			✔
LIGHT GREEN	1/3	T1596	Search Open Technical Databases	Reconnaissance	✔		
LIGHT GREEN	1/3	T1606	Forge Web Credentials	Credential Access		✔	
LIGHT GREEN	1/3	T1606.001	Web Cookies			✔	
LIGHT GREEN	1/3	T1606.002	SAML Tokens			✔	
LIGHT GREEN	1/3	T1608	Stage Capabilities	Resource Development			✔
LIGHT GREEN	1/3	T1608.001	Upload Malware				✔
LIGHT GREEN	1/3	T1621	Multi-Factor Authentication Request Generation	Credential Access		✔	
LIGHT GREEN	1/3	T1649	Steal or Forge Authentication Certificates	Credential Access		✔	
LIGHT GREEN	1/3	T1651	Cloud Administration Command	Execution		✔	
LIGHT GREEN	1/3	T1665	Hide Infrastructure	Command and Control		✔	
LIGHT GREEN	1/3	T1669	Wi-Fi Networks	Initial Access	✔		
LIGHT GREEN	1/3	T1680	Local Storage Discovery	Discovery		✔	
LIGHT GREEN	1/3	T1684	Social Engineering	Stealth	✔		
LIGHT GREEN	1/3	T1684.001	Impersonation		✔		